Brick Atlas: From Static Repair to Autonomous Recovery in Brick Assembly Supplementary Material

Anonymous CVPR submission

Paper ID Draft

A. Current Release and Evidence

This supplement describes the release of 29 October 2026: 310 runnable object configurations, 1,600 static questions, and 3,085 dynamic questions. The collection comprises 263 admitted community configurations and 47 retained designed whole objects. All 4,685 published questions have verified reference executions. These counts supersede earlier manuscript snapshots; they do not convert historical model pilots into results on the current interface.

The release distinguishes three layers: acquired sources, admitted executable configurations, and questions instantiated on them. The source pool has 437 authored objects from 338 original sets, including 99 authored alternatives. Of the 310 runnable configurations, 57 are complete builds and 253 are finishing tasks. A further 174 acquired sources are not admitted to this release. An acquired file is not a runnable benchmark configuration.

The authoritative files are the release manifest, object JSONL, static and dynamic question JSONL, and evaluator-private certificates. The paper snapshots these dependencies in `evidence/release-dataset.json`; selected figure states and case IDs are recorded in `evidence/release-render-states.json` and `evidence/release-figure-selection.json`. Figures are publication exports of exact geometry. Their cameras and lighting do not replace the model's native observations.

A.1. Historical records and exclusions

The historical browser preserves 8,997 catalogue records across versions. Only 4,685 are current checked questions; the remainder includes 1,983 retired-source or human-selection records, 2,265 historical records not rechecked, 31 withheld reference/visibility conditions, and 33 pending historical records. This is an archival inventory, not a claim of 8,997 unique valid tasks. Earlier counts such as 5,720 distinct records belong to their own contracts and selected populations.

The 31 withheld designed-object conditions remain in the failure ledger: eighteen do not reach every configured incident under the current reference, twelve lack a verified effective change, and one static defect is invisible in all three default views. The current 1,605 retained designed-object questions are the successful subset. One community camper candidate is separately withheld because its original source color cannot be matched without guessing. User-retired buildings and minifigure-containing sources are excluded from admission. No exception is counted as a model failure.

A.2. Source grouping and overlap

There are 234 source groups among runnable objects. Authored alternate builds remain in the same group as their original set. Development, validation, and test labels separate source groups, but all current sources have been used during authoring. They therefore do not certify an untouched test distribution. Future external-source evaluation must declare newly acquired source IDs, keep them out of prompt and engine tuning, and report failures in qualification before sampling tasks.

B. Construction and Admission Procedure

B.1. Representing a complete source

Community certificates retain source-instance indices, part numbers, colors, and original transforms. Instances are assigned to the operable target, fixed supported context, or conservative background context. The admission check rereads the original file and requires the union of represented indices to equal the complete source instance set. Part aliases are accepted only when resolved part definitions are identical. Raw matrices are checked numerically with a fixed tolerance.

For small complete objects, the target includes every object part and construction begins on a white studded baseplate. For larger sources, the start supplies the body and the scope describes a finishing region. A whole source may contain many more parts than the task operates on. The illustrated gallery retains this body so the remaining work

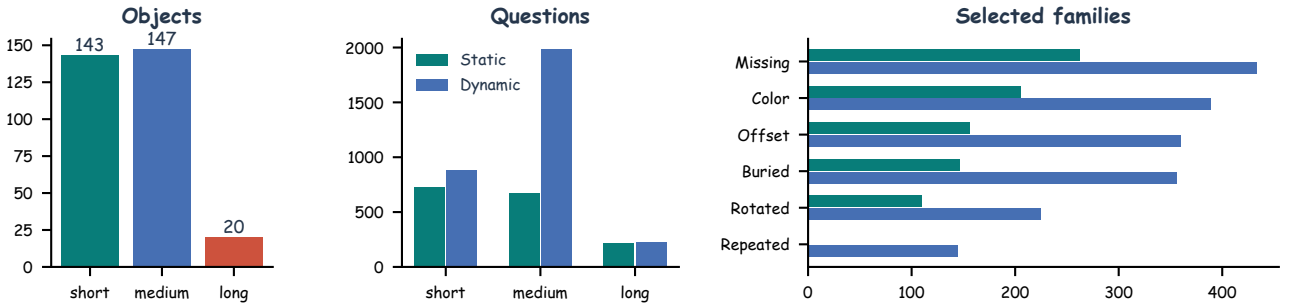


Figure S1. **Coverage without inflated independence.** Left: runnable objects by length. Middle: published static and dynamic questions by length. Right: selected family counts split by mode. These are dataset statistics, not measured difficulty or model performance. Detailed family and source tables accompany the release.

Table S1. **Current runnable dataset.** S0/S1 are correct-stage acceptance/static repair; D0/D1/D2 are normal construction/one incident/repeated incidents. Source acquisitions and historical records do not enter these counts.

Scope	Objects	S0	S1	D0	D1	D2	Questions
Complete build	57	41	141	57	1,466	28	1,733
Finishing task	253	253	1,165	253	1,165	116	2,952
Total	310	294	1,306	310	2,631	144	4,685

Table S2. **Length descriptors, not calibrated difficulty.** Active parts exclude fixed context.

Length	Objects	Static	Dynamic
short (3–7)	143	721	884
medium (8–15)	147	667	1980
long (16+)	20	212	221

Table S3. **Published question families.** One catalogue family per question; this table is not an exclusive physical-error ontology.

Family	Static	Dynamic	Total
buried-repair	146	356	502
compound	0	122	122
control	294	310	604
early-cover	57	57	114
extra-layer	80	229	309
left-right-swap	24	24	48
missing	262	433	695
multiple	0	144	144
offset	156	360	516
repeat-omission	136	136	272
repeated-step	78	78	156
rotated	110	225	335
swapped-positions	0	11	11
wrong-color	205	389	594
wrong-part	52	211	263

has visible context. Reviewers can choose module focus, whole-object focus, and several angles.

The source pool is audited for minifigure parts. An additional exact structural duplicate check ignores color and

translation and normalizes axis-aligned rotations. Its absence of duplicates does not prove that two cars or houses look sufficiently different. Human assessment of meaningful diversity is a separate requirement.

B.2. Static and dynamic variants

The community static bank selects a normal stage and distinct faulty states, with two to six questions per admitted object. S0 is a correct intermediate acceptance target, not an invitation to finish all subsequent instructions. S1 uses a supplied faulty state and its stage-specific target. Every question has its own initial and target image paths, so two variants cannot silently display the same stale images.

Dynamic D0 starts normal construction. D1 and D2 retain the same target while specifying one or multiple potential external changes. A reference witness verifies a feasible continuation after each admitted incident. A state-based trigger may defer when an eligible intervention is unavailable. Source diversity and feasible repair determine admission; the generator does not force every error category onto every object.

The family field is a catalogue grouping, not a full causal taxonomy. Missing, color, position, part, and orientation describe discrepancies. Buried repair and early cover describe access demands. Repeated events describe episode structure. Compound errors can involve more than one physical discrepancy. Analyses should preserve these distinctions rather than interpret the table as mutually independent human-error mechanisms.

B.3. Reference order and reference effort

The scorer compares physical placement multisets and final constraints. It does not compare the action sequence with a single demonstration. The checker searches adjacent independent swaps in normal reference sequences and verifies successful alternative trajectories for 251 of the 263 community configurations. The remaining twelve may still have other legal orders outside this bounded search.

References support solvability checks and an effort baseline. They are not certified global shortest paths. Removing a correct cover can be necessary; removing an unrelated column can be unnecessary. Human GT review should inspect this distinction. A model taking fewer actions than the recorded reference can be correct and can have a reference-relative effort below one.

C. Actual Input and Output Contracts

C.1. Dynamic initial request

The Arena initial request contains the task description, starting-state description, sole target contract, neutral physical rules, decision cap, camera metadata, number of inventory and instruction pages, anonymous action handles, and six tool schemas. It does not contain initial RGB automatically. The protocol identifier is `brick-atlas-tool-arena/3`.

All subsequent replies contain one tool call. The following are syntax examples; identifiers must come from the episode's returned inventory and placement lists. They are separate decisions, not a batch action list.

```
{"tool":"observe", "args":{
"view":"angled", "rotateDegrees":90,
"show":["current","target"]}}

{"tool":"inspect", "args":{
"partId":"A00001", "page":1}}

{"tool":"attach", "args":{
"partId":"A00001",
"candidate":"C00001"}}

{"tool":"detach", "args":{
"partId":"A00001"}}

{"tool":"finish", "args":{}}
```

The executable artifact uses standard ASCII JSON quotes. Typeset examples here illustrate the fields; exact copyable requests are in the dataset interface guide and Arena log.

C.2. Observation semantics

The angled camera accepts relative rotations of -90, 0, 90, or 180 degrees. A top view sets elevation to 90 degrees. Each requested current or target scene is 800 by 600 pixels, with matched camera and scope. The default focus is the operable region; whole focus includes all context. The model can request current, target, and inventory images independently in one observation call.

Inventory contact sheets paginate at twelve parts. Live installed/available status accompanies a requested inventory. Candidate inspection displays possible placements in the actual scene, with the candidate visually outlined. These cards do not label legal or target positions. An instruction-page request returns one optional manual step, marked as a reference rather than the acceptance target. A recall request retrieves previously requested images and their original decision number; it does not observe the current state.

After attach or detach, feedback describes acceptance, physical rejection, resource rejection, or invalid parameters. No fresh scene image is attached automatically. The model can continue building without looking or spend a decision to observe. The spectator page's three-angle images are generated for review and are not implicitly delivered to the model.

C.3. Static adapter

At each decision, static tasks provide three matched current/target views and an inventory contact sheet. Responses use `type`, rather than the dynamic envelope's `tool` and

`args`. For example, `{"type":"finish"}` submits acceptance; inspect, attach, and detach expose the same underlying scene semantics. The different sensing policies are versioned. Claims that static and dynamic scores isolate only an intervention require an additional matched-sensor experiment.

C.4. Public and evaluator-private data

The public model request may reveal anonymous inventory and placement handles, optional instructions, requested RGB, and neutral action feedback. It must not reveal error-family labels, hidden schedules, reference actions, private goal dictionaries, event counts, or evaluator success signals while the episode is active. Human review pages can show GT separately; they are explicitly answer-bearing surfaces. Trace exports record what the model actually saw rather than substituting spectator images.

D. Metrics and Denominators

D.1. Exact completion and partial state quality

Placement signatures contain part type, color, and world pose, with multiplicities for interchangeable parts. Fixed context is not included in operable-placement F1. Exact success requires explicit finish, equality with the target signature, and any final connection constraints. An accepted off-target move is legal but may need later correction; it is not an automatic failure.

For static S1, report exact SR, full-placement F1, initial and final residual, error reduction, originally missing target content recovered, original excess cleared, and final damage to initially correct placements. Residual is the number of missing and excess placements. It is not shortest repair distance. Error reduction is undefined for an initially exact state and can be negative after deterioration. S0 separately reports correct acceptance and whether any physical move was accepted.

D.2. Dynamic normal and incident runs

BCR uses valid D0 runs. ICR is valid completion of incident-assigned runs, irrespective of whether the incident was reached. Trigger coverage retains scheduled and applied counts. The primary normal-qualified recovery measure further requires a successful matched normal run, a valid terminal status, complete incident exposure, and verified effective change. Its success field remains null when ineligible.

The matched normal and incident runs share the task, evaluated model, input contract, and budget, but are independently executed. This supports a descriptive conditional recovery rate. It does not support a causal completion drop. A strict paired estimator requires real evidence of a cloned prefix, including messages and already spent budget. The scorer supports that distinction; a completed shared-prefix agent study is not supplied by this release.

D.3. Event windows

An event window starts after the incident and ends immediately before the next incident. Lost correct content is measured relative to the before-event state and the final target; introduced excess is measured relative to pre-existing excess. Recovery can coexist with forward progress. It need not reproduce the old prefix exactly.

First recovery is counted in subsequent decisions, including observations and invalid actions. Retention requires at least two follow-up decisions. Observed relapse fails retention; insufficient follow-up yields N/A. Pure access obstructions without a new placement defect use completion-related evaluation instead of declaring instantaneous recovery. For repeated incidents, partial exposure and incomplete later recovery remain explicit.

D.4. Failures and aggregation

The terminal statuses eligible for ability evaluation are completed and model-budget-exhausted. A wrong explicit finish or exhausted model budget is failure. Service error, harness error, and external stop are separate technical outcomes. Format errors and schema errors are distinct from collisions, inaccessible paths, wrong placements, and out-of-stock responses. Recovery cannot be used as a direct detection label.

Every model table should provide planned attempts, valid attempts, event exposure, and metric eligibility. Report task-micro and object/source-macro results. Source-clustered intervals preserve correlated variants. A single run per condition describes that run and the sampled task set; it does not quantify within-model run variance. Cost totals include observation images, instruction pages, recalls, retries, prompt/history tokens, generated tokens, and local computation when applicable.

E. Verification and Remaining Checks

Reference replay verifies all published questions and source consistency. The image audit verifies 17,618 scene assets and current/target agreement in size. Normal controls match exactly. Community static defects differ in at least one exported view. This nonempty-pixel test does not establish human salience, nor does a finite set of views prove that every adaptive event is visible at every runtime location.

Six actual ToolArena scripted episodes cover three objects with normal and incident conditions: the complete cow, a truck finishing task, and a longer building task. They exercise real tool envelopes, requested images, action feedback, terminal scoring, and normal-qualified recovery. These are privileged witnesses; they use reference knowledge and are never reported as learned-agent successes. The

final check also runs metric and contract regressions for alternative orders, correct intermediate acceptance, untriggered events, incomplete exposure, and technical failures.

Independent human checks must distinguish displayed-GT review from blind operation. Reviewers first assess whether the target and operable region are clear, then whether the proposed repair changes the right parts without unnecessary collateral work. Blind operators receive only model-visible evidence and controls. Their outcomes establish usability and ability, while GT agreement only supports annotation quality. No copied agreement from one reviewer supplies another independent observation.

Independent mechanical checking should examine buried parts, interlocking geometry, conservative background collisions, extraction paths, and whether removing a support invalidates other parts. The present engine uses fixtures and does not simulate gravitational collapse or deformation. No program-only audit is labeled independent physical validation.

F. Archived API Examples and Reproducible Comparison

The main-paper examples are selected Arena v2 API episodes with successful independent normal controls, not current-release scores. The car has 21 decisions, nine observations, seven accepted moves, three format failures, one collision rejection, and a correct finish. The chair has 30 decisions, seven observations, seventeen accepted moves, five physical rejections, and a wrong finish (TP=5, FP=1, FN=2). Different objects and selected outcomes do not establish a model ranking.

The publication exporter replays every recorded action, checking before state, own-action state, post-incident state, feedback, and new event indices. It also checks the final outcome. For each GT row, the actual prefix is retained through the external removal. A privileged continuation restores the affected part, executes a legally checked target completion, and finishes. This is a feasible reference from the same faulty state, not a causal pair of two learned continuations. Displayed decision numbers locate selected frames; the full ledger preserves all omitted observations, invalid requests, and manipulations.

Source trace hashes, selected state hashes, both normal-control IDs, and GT continuation actions are in `evidence/arena-traces.json`. The accompanying Chinese trace guide lists every public tool call and feedback. A statement about observation uses requested-image timestamps; a statement about repair uses actual state changes. Neither a camera request nor a verbal claim is independently scored as successful detection.

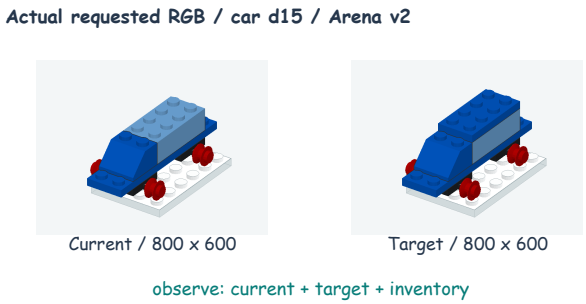


Figure S2. **Actual model evidence.** Original requested RGB from the car's decision 15. The same call also requested inventory; the response was OBSERVED at azimuth 35°, elevation 30°. Hidden event labels and GT plans are absent. High-resolution trajectory images elsewhere are spectator exports, not these input bytes.

F.1. Complete pilot ledger and limits of attribution

The archival audit includes sixteen normal attempts and three incident attempts. For a per-condition view it selects the latest recorded normal attempt for each of four models and three objects, plus all three incidents: fifteen conditions, twelve valid. The other four attempts ended in service interruptions and remain in the all-attempt ledger. The three latest Qwen conditions also ended in service errors. None supplies an ability score. This is a descriptive reanalysis, not a retrospectively registered experiment.

Invalid requests concern the output contract. Unregistered pose handles concern the candidate vocabulary; the historical summary grouped them under physical rejections, so this analysis explicitly reclassifies them without rewriting raw logs. NO_CONNECTION, ENDPOINT_COLLISION, and PATH_BLOCKED record different geometric constraints. A rejected action leaves the arrangement unchanged and is not a brick actually installed in the wrong place. Endpoint residual, wrong finish, and decision exhaustion are separate outcomes; none uniquely identifies a perceptual or reasoning cause.

G. Additional Diagnostic Protocols

G.1. Observation, progress, and stopping

For an effective event after decision τ, define fresh-view delay as $\min\{t - \tau : t > \tau,\ a_t \text{ requests current RGB}\}$. A recall of an old image is not fresh. If no current view is requested before termination, record the observed window length and a censored indicator rather than inventing a numeric latency. Camera availability does not guarantee that the affected part is visible; visibility qualification is a separate field. This metric measures acquisition of evidence, not comprehension.

Recovery@k checks event recovery within k subsequent decisions on valid, exposed, normal-qualified windows. Ter-

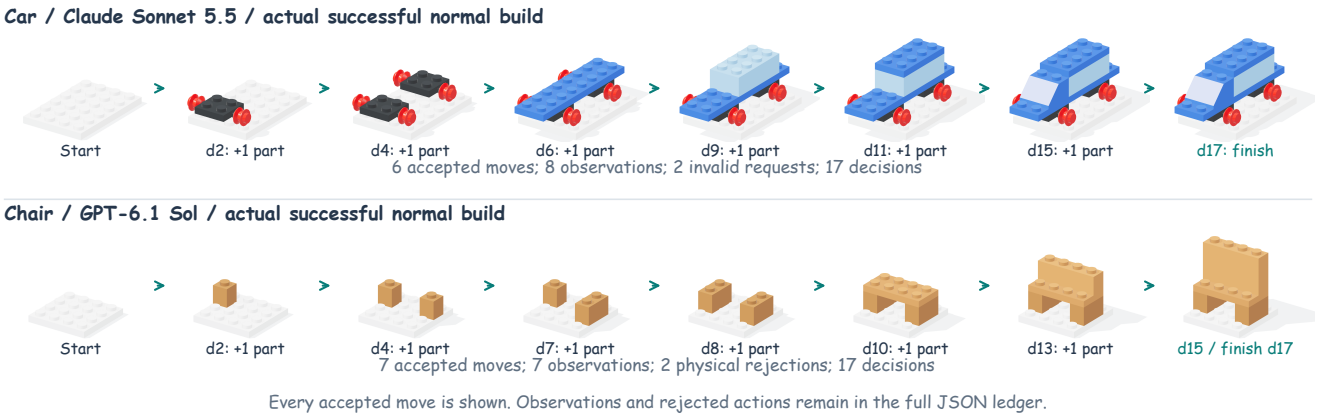


Figure S3. **Actual successful assembly, one accepted move at a time.** Car: all six accepted placements; chair: all seven. Both finish at decision 17. The chair's last frame combines its final placement at 15 and unchanged finish at 17. Interleaved observations and rejected requests are preserved in the public decision ledger. These selected historical v2 successes demonstrate genuine model-built progress, not a success rate for the current release.

Table S4. **All latest archived pilot conditions (Arena v2).** Fmt: invalid request; ID: unregistered pose handle; Conn/Coll/Path: rejected connection, endpoint collision, or extraction path. Counts are actions, not independent cases. Service errors have no ability outcome. Four earlier service-interrupted attempts remain in the all-attempt ledger (19 total). These are not v3 benchmark results.

Model / object	Task	Outcome	Fmt	ID	Conn	Coll	Path	Moves
Qwen3-VL-235B / city-car	D0	service	—	—	—	—	—	—
Qwen3-VL-235B / crane	D0	service	—	—	—	—	—	—
GPT-6.1 Sol / chair	D0	success	0	0	1	1	0	7
Qwen3-VL-235B / chair	D0	service	—	—	—	—	—	—
Claude Sonnet 5.5 / chair	D0	success	6	0	1	0	0	7
Gemini 2.5 Flash / chair	D0	budget	0	0	3	11	0	5
Gemini 2.5 Flash / city-car	D0	budget	0	2	2	12	0	3
Gemini 2.5 Flash / crane	D0	budget	0	1	12	0	0	2
GPT-6.1 Sol / city-car	D0	budget	0	0	2	2	4	16
GPT-6.1 Sol / crane	D0	budget	0	0	13	3	0	8
Claude Sonnet 5.5 / city-car	D0	success	2	0	0	0	0	6
Claude Sonnet 5.5 / crane	D0	budget	1	0	12	0	0	5
GPT-6.1 Sol / chair	D1	wrong finish	0	0	1	3	1	17
Claude Sonnet 5.5 / chair	D1	wrong finish	3	0	0	0	1	9
Claude Sonnet 5.5 / city-car	D1	success	3	0	0	1	0	7

minal states persist through the horizon: an early unrepaired finish is a failure, not an excluded short run. A new incident or technical stop before k censors the window and must be counted separately. Report end-of-window recovery and recurrence alongside first recovery; repair can occur without another view.

Let q_t be operable-placement F1 after decision t. Mean and peak F1 accompany final success and episode length; a high average cannot excuse early termination. A stall spans accepted moves without reduced placement residual; observations are excluded and external edits split segments. Access work can count as a stall, so report access type and final recovery. Wrong-finish rate divides unacceptable explicit finishes by all explicit finishes, with counts. These remain planned trace analyses.

G.2. Independence, fairness, and controlled assistance

E24 tests transformations with known expected behavior. Reordering independent actions and swapping physically interchangeable handles preserve endpoint acceptance; deliberately wrong endpoints do not. Reversible access work changes effort while preserving final damage. For action adjudication (E25), independently valid and invalid actions form separate denominators: false rejection is engine rejection among independently valid actions, and false acceptance is engine acceptance among independently invalid ac-

tions. Reviewers must see geometry and proposed motion before seeing the engine verdict. Unresolved judgments remain outside both denominators and retain their own count.

E21 discloses progressively more information: fault time, affected part, symbolic state, or repair plan. Such assistance changes the task and cannot uniquely diagnose internal reasoning. E20 compares simple policies with learned agents; privileged witnesses remain feasibility controls. New aggregate results stay blank until the common-task study is executed.

G.3. Matched-state transfer and source uncertainty

E29 compares a fresh solver with the continuing agent at the same realized fault, crossed with active and scheduled sensing. Match target, inventory, candidates, allowed views, and remaining decisions. The continuing agent retains its history; a separate equal-total-budget analysis includes reaching the checkpoint. This controls task differences before attributing a performance gap to autonomy.

E23 changes incident stage, magnitude, or spacing and rechecks feasibility. E27 uses genuinely unused sources. E28 bootstraps source groups with their correlated variants. With one run per condition, intervals describe source sampling, not repeated-generation variance. Report eligible coverage and common-qualified-object sensitivity alongside rankings.

H. Error Families and Mechanism Probes

A task's defect is a change in the world. A model error is observed behavior under that change. The two should not be conflated: the same missing part can be overlooked, localized incorrectly, reached by an invalid motion, deliberately postponed, or left unrepaired after budget exhaustion. The following demands are hypotheses to test with controlled probes, not internal-cause annotations.

E32 uses separate cloned checkpoints to test neutral scene description, discrepancy localization, legal-pose selection, access ordering, supplied-plan execution, state updating, and finish judgment. Primary episodes receive none of the probe answers. A supplied feasible plan that still fails exposes an execution/interface bottleneck under that assistance; it does not prove the original failure had only that cause. Successful textual description with failed repair narrows the hypothesis but still permits grounding, ordering, and stopping errors.

I. Text, Image, and Combined Evidence Protocol

The experimental axis crosses S0/S1/D0/D1/D2 with T/I/TI without creating new underlying questions. Modality is a run configuration, never an additional object or question count. The same candidate handles and legal action domain remain available. Target part descriptions must omit actor-to-target assignment when interchangeable identities are not part of the acceptance contract; otherwise text would hand out the placement solution.

T uses neutral structured records: part type, color, position in declared units, orientation, and installed/available inventory status. Current and target are separate records with identical schemas. I uses native current/target views plus visual inventory and candidate cards; stock status appears in those cards, not as a separate textual state list. TI returns both channels generated from the same state and timestamp. Tool syntax and neutral feedback remain textual in every condition. Optional instructions and hypothetical placements must also follow the condition; disabling them in only one arm confounds the comparison.

Static evidence is refreshed at each decision as in the existing static adapter. Dynamic initial requests contain only the common contract, handles, and budgets. An explicit observation fetches current/target/stock in the assigned representation; no move silently adds a fresh state. An incident changes the world but does not update the conversation until a requested observation. Recall remains historical. If a hidden edit changes stock, a fresh stock query is legitimate evidence; an unsolicited stock refresh would leak the event.

Native interfaces versus matched facts. Exact text is an oracle state description relative to native RGB: it can expose invisible parts and precise poses. The native T/I/TI experiment measures the utility of these interfaces, not equal-information perception. The matched-fact control instead uses one neutral ledger, serialized as text, rendered as readable image cards, or both. All facts, including hidden geometry and stock, are identical. This is a representation/OCR control and cannot replace the native 3D task. A view-limited caption alternative would need a separately validated visibility/uncertainty policy and is not silently assumed here.

Validity and reporting. Check T/I/TI against the same underlying state, target, candidate domain, and timestamp before model use. Inspect provider payloads to ensure T contains no images and strict I contains no machine-readable scene/stock ledger outside images. Neutral API schemas and image labels are allowed. No channel reveals incident metadata, reference actions, correctness-colored candidates, per-part goal assignment, or precomputed target differences. Modality failures are harness issues until resolved. Report payload size, tokens, images, query decisions, and cost without pretending text and image tokens are equal.

E11 reports S1 exact repair, F1, damage, and S0 false repair by modality and family. E31 reports construction and normal-qualified one/repeated-event recovery, plus cov-

Table S5. **Additional evidence required for trustworthy interpretation (reserved).** Each result cell is unmeasured. Planned experiments E20–E32 have complete result sheets in the supplement.

Check	Primary quantity	Cases	Result
E20: simple-policy controls	Completion and observation cost	—	—
E22: active observation	Fresh-view delay; recovery within 4/8 decisions	—	—
E24: alternative-order fairness	Verdict agreement after legal transformations	—	—
E25: independent action review	False acceptance / false rejection	—	—
E27: unused-source evaluation	Source-macro completion and recovery	—	—
E29: matched faulty states	Fresh/continuing agent $\times$ sensing policy	—	—
E30: trajectory and stopping	Mean/peak F1; stalls; wrong finish	—	—

Table S6. **Different errors, different demands.** All named families occur in the current release. Repeated incidents are a scheduling family, and controls are not errors. A demand is not an exclusive cognitive diagnosis.

Family	Main observable demand	Separate evidence needed
Missing	Find absent target content	Count and localize before repair
Extra layer	Distinguish surplus from required support	Select excess without damaging correct content
Wrong color	Bind an attribute to the correct location	Color/handle description then repair
Wrong part	Distinguish shape and part identity	Type recognition, candidate grounding
Offset	Align translations and connections	Pose choice then legal execution
Rotated	Resolve orientation and contact direction	Rotation probe, collision/path verdict
Left-right swap	Bind identity to side in a shared frame	Label both positions, then swap legally
Swapped positions	Reconcile two occupied placements	Multi-part correspondence and ordering
Repeat omission	Count separate repeated instances	Per-instance count and localization
Repeated step	Identify duplicated installed content	Distinguish repetition from required symmetry
Early cover	Preserve access to unfinished work	Open-cover / finish / restore sequence
Buried repair	Reach an interior fault	Access ordering, collateral work, restoration
Compound	Reconcile interacting edits	Component residual and complete recovery
Multiple events	Update repeatedly without resetting	Event-wise recovery and retained repair
Control	Accept an already correct state	False-repair and correct-finish rates

erage and observation use. Each mode requires its own successful normal control; also compare the intersection of qualified objects to expose selection effects. E29 uses identical saved faulty states and remaining budgets, with history and sensing policy controlled, within each modality. Formal results remain blank: the release currently implements native RGB plus textual stock status; the three strict adapters and their payload audits are pending.

J. Planned Experiments and Empty Result Sheets

Thirty-two result sheets remain unmeasured; dashes are not zeros and JSON result fields are null. Pin exact model versions, inputs, decoding, and budgets before one run per condition. Tables identify protocols requiring further integration. No new API calls were made.

Table S7. **E1: static repair (unmeasured).** Implemented. Separate whole-object and finishing stages; SR/F1/ER/DR use S1 and FR uses S0. Dashes mean not run, never zero.

Model	SR	F1	ER	DR	S0 FR	Valid N
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table S8. **E2: normal assembly and recovery (unmeasured).** Implemented normal-qualified recovery uses independent matched runs, not causal forks. Dashes mean not run, never zero.

Model	BCR	D1 NQR	D2 NQR	ICR	TR	Eligible N
GPT-6.1 Sol	—	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—	—

Table S9. **E3: strict paired outcomes (unmeasured).** Planned shared-prefix study. Only genuine cloned-prefix pairs enter CRR or the paired difference. Dashes mean not run, never zero.

Model	Pairs	Control SR	Fault SR	CRR	ΔSR
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S10. **E4: evidence ablations (unmeasured).** Planned matched-sensor comparison; change one factor and report actual images. Dashes mean not run, never zero.

Condition	SR	BCR	NQR	Queries	USD
Full tools	—	—	—	—	—
No manual pages	—	—	—	—	—
One requested view	—	—	—	—	—
No recall	—	—	—	—	—
Scheduled observations	—	—	—	—	—

Table S11. **E5: static families (unmeasured).** Implemented breakdown; fit only families with static support; do not pool controls. Dashes mean not run, never zero.

Family	SR	F1	ER	DR	Valid N
buried-repair	—	—	—	—	—
compound	—	—	—	—	—
control	—	—	—	—	—
early-cover	—	—	—	—	—
extra-layer	—	—	—	—	—
left-right-swap	—	—	—	—	—
missing	—	—	—	—	—
multiple	—	—	—	—	—
offset	—	—	—	—	—
repeat-omission	—	—	—	—	—
repeated-step	—	—	—	—	—
rotated	—	—	—	—	—
swapped-positions	—	—	—	—	—
wrong-color	—	—	—	—	—
wrong-part	—	—	—	—	—

Table S12. **E6: dynamic families (unmeasured).** Implemented breakdown; do not infer detection from completion. Dashes mean not run, never zero.

Family	ICR	TR	NQR	Eligible N
buried-repair	—	—	—	—
compound	—	—	—	—
control	—	—	—	—
early-cover	—	—	—	—
extra-layer	—	—	—	—
left-right-swap	—	—	—	—
missing	—	—	—	—
multiple	—	—	—	—
offset	—	—	—	—
repeat-omission	—	—	—	—
repeated-step	—	—	—	—
rotated	—	—	—	—
swapped-positions	—	—	—	—
wrong-color	—	—	—	—
wrong-part	—	—	—	—

Table S13. **E7: preservation and partial repair (unmeasured).** Implemented; temporary removal is effort, not final damage. Dashes mean not run, never zero.

Model	F1	Missing fixed	Excess cleared	Damage	Moves
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S14. **E8: length and access (unmeasured).** Length bands are current. Access calibration and empirical difficulty require independent evidence. Dashes mean not run, never zero.

Stratum	Objects	SR	BCR	NQR	Moves
Short (3–7)	—	—	—	—	—
Medium (8–15)	—	—	—	—	—
Long (16+)	—	—	—	—	—
Accessible defect	—	—	—	—	—
Buried repair	—	—	—	—	—

Table S15. **E9: repeated incidents (unmeasured).** Implemented event windows end before the next external incident; short retention windows are N/A. Dashes mean not run, never zero.

Model	All-event SR	Exposure	Recovered	Retention	Eligible N
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S16. **E10: supplied checkpoints (unmeasured).** Planned separate bank; system-supplied progress is not model-built progress. Dashes mean not run, never zero.

Model	SR	Repair@2	Repair@4	Repair@8	Valid N
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S17. **E11: static input representation (unmeasured).** Planned adapters. First three compare native interfaces with unequal perceptual information; last three encode identical facts and test representation/OCR. Keep family and visibility strata; never pool the two studies. Dashes mean not run, never zero.

Input	S1 SR	F1	S0 FR	Damage	Valid N
T: structured text	—	—	—	—	—
I: native scene images	—	—	—	—	—
TI: both	—	—	—	—	—
T: matched fact ledger	—	—	—	—	—
I: rendered fact cards	—	—	—	—	—
TI: matched facts in both	—	—	—	—	—

Table S18. **E12: localization interface (unmeasured).** Current handles are implemented; coordinate variants require matched action domains. Dashes mean not run, never zero.

Output	SR	BCR	NQR	Schema err.	Phys. reject
Candidate handle	—	—	—	—	—
Discrete coordinates	—	—	—	—	—
Continuous coordinates	—	—	—	—	—

Table S19. **E13: independent detection (unmeasured).** Planned independent snapshot bank; no label inferred from episode failure. Dashes mean not run, never zero.

Model	Detection	Localization	False alarm	Visible N
GPT-6.1 Sol	—	—	—	—
Claude Sonnet 5.5	—	—	—	—
Gemini 3.8 Flash	—	—	—	—
Claude Haiku 5.5	—	—	—	—
Qwen3.5-9B	—	—	—	—
Qwen3.5-27B	—	—	—	—
InternVL3.5-8B	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—
Gemma 3-12B	—	—	—	—

Table S20. **E14: repair of another agent state (unmeasured).** Planned shared failed-state bank with matched checkpoints and no hidden reference access. Dashes mean not run, never zero.

Repairer	Own state SR	Other state SR	Residual	Valid N
GPT-6.1 Sol	—	—	—	—
Claude Sonnet 5.5	—	—	—	—
Gemini 3.8 Flash	—	—	—	—
Claude Haiku 5.5	—	—	—	—
Qwen3.5-9B	—	—	—	—
Qwen3.5-27B	—	—	—	—
InternVL3.5-8B	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—
Gemma 3-12B	—	—	—	—

Table S21. **E15: observation robustness (unmeasured).** Planned controlled visual variants; recheck visibility and legality. Dashes mean not run, never zero.

Condition	SR	BCR	NQR	Queries
Matched native views	—	—	—	—
Higher resolution	—	—	—	—
Restricted camera	—	—	—	—
Occluded initial view	—	—	—	—
Changed handle seed	—	—	—	—

Table S22. **E16: planning and budget (unmeasured).** Planned study; charge all decisions and queries, and change one budget at a time. Dashes mean not run, never zero.

Condition	SR	BCR	NQR	Decisions	USD
Default decision cap	—	—	—	—	—
Half decision cap	—	—	—	—	—
Double decision cap	—	—	—	—	—
Observation schedule	—	—	—	—	—

Table S23. **E17: effort and cost (unmeasured).** Implemented logging. Reference is feasible, not optimal; retry costs remain included. Dashes mean not run, never zero.

Model	Moves/ref.	Observe	Recall	Tokens	USD
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S24. **E18: independent human study (unmeasured).** Planned. GT review is not blind ability evidence; responses must be independent. Dashes mean not run, never zero.

Panel	N	Completion	Exposure	Agreement	Time
Blind normal assembly	—	—	—	—	—
Blind incident recovery	—	—	—	—	—
GT correctness review	—	—	—	—	—
Physical feasibility review	—	—	—	—	—

Table S25. **E19: validity ledger (unmeasured).** Implemented status separation. Planned and valid denominators are both reported. Dashes mean not run, never zero.

Model	Planned	Valid	Service err.	Format err.	Untriggered
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S26. **E20: reference and simple policies (unmeasured).** Planned common-task controls. Witness uses private state and is a feasibility check, never a fair learned-agent competitor. All aggregate cells remain unmeasured. Dashes mean not run, never zero.

Policy	S1 SR	D0 BCR	D1 NQR	Moves	Queries
No-op / finish	—	—	—	—	—
Uniform random handles	—	—	—	—	—
Manual-only executor	—	—	—	—	—
Periodic visual checker	—	—	—	—	—
Learned full-tools agent	—	—	—	—	—
Privileged feasible witness	—	—	—	—	—

Table S27. **E21: assistance ladder (unmeasured).** Planned diagnostics with increasing privileged information; report assistance explicitly. These conditions cannot replace the public-input leaderboard. Dashes mean not run, never zero.

Assistance	SR	NQR	Damage	Moves	Eligible N
Ordinary RGB tools	—	—	—	—	—
Fault time disclosed	—	—	—	—	—
Fault part disclosed	—	—	—	—	—
Exact symbolic state	—	—	—	—	—
Feasible repair plan	—	—	—	—	—

Table S28. **E22: observing and recovering (unmeasured).** Planned postprocessing: delay to first fresh current image after an event, not detection time. Recovery@k retains early terminal outcomes and censors a subsequent event before k; report horizon coverage. Dashes mean not run, never zero.

Model	Observe delay	Recovery delay	Repair@4	Repair@8	Visible N
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S29. **E23: incident stage and magnitude (unmeasured).** Planned controlled incident variants. Stage is measured by actual correct installed fraction; requalify solvability after every change. Dashes mean not run, never zero.

Condition	TR	NQR	Residual	Moves	Eligible N
Early / one changed part	—	—	—	—	—
Middle / one changed part	—	—	—	—	—
Late / one changed part	—	—	—	—	—
Matched stage / two parts	—	—	—	—	—
Short gap / two events	—	—	—	—	—
Long gap / two events	—	—	—	—	—

Table S30. **E24: scorer invariance (unmeasured).** Planned scored paired audit; known existing checks do not populate a new aggregate automatically. Preserve final geometry and constraints for invariant pairs; wrong finals must be rejected. Dashes mean not run, never zero.

Transformation	Pairs	Same verdict	Delta F1	Mismatch N
Independent action swap	—	—	—	—
Identical-part handle swap	—	—	—	—
Legal remove and replace	—	—	—	—
Correct cover access	—	—	—	—
Wrong final arrangement	—	—	—	—

Table S31. **E25: independent action adjudication (unmeasured).** Planned independent expert or separate-engine verdicts. False acceptance divides by independently invalid actions; false rejection divides by independently valid actions. Unresolved cases remain separate. Dashes mean not run, never zero.

Action stratum	Actions	False accept	False reject	Agreement
Accessible insertion	—	—	—	—
Buried extraction	—	—	—	—
Tight contact / interlock	—	—	—	—
Supporting-part removal	—	—	—	—
Conservative background	—	—	—	—

Table S32. **E26: evidence usability (unmeasured).** Planned independent review with only model-visible evidence. Count recoverably occluded and never-visible cases separately; agreement after showing GT is not blind performance. Dashes mean not run, never zero.

Panel	Cases	Target clear	Fault visible	Action clear	Agreement
Initial supplied view	—	—	—	—	—
After permitted rotation	—	—	—	—	—
Finishing-region focus	—	—	—	—	—
Repeated identical parts	—	—	—	—	—
Blind tool operation	—	—	—	—	—

Table S33. **E27: source and object coverage (unmeasured).** Planned external-source study; newly acquired sources must be absent from authoring and tuning. Current test labels alone do not establish novelty. Dashes mean not run, never zero.

Source stratum	Objects	SR	BCR	NQR	Eligible N
Current authoring-exposed	—	—	—	—	—
New source / known family	—	—	—	—	—
New source / new family	—	—	—	—	—
Complete objects	—	—	—	—	—
Contextual finishing tasks	—	—	—	—	—

Table S34. **E28: aggregation and eligibility (unmeasured).** Planned source-cluster bootstrap with one run per condition. Intervals describe source sampling, not model sampling variance; all-attempt view reports technical exclusions without scoring them as ability failures. Dashes mean not run, never zero.

Summary	SR	BCR	NQR	CI width	Coverage
Task micro	—	—	—	—	—
Object macro	—	—	—	—	—
Source macro	—	—	—	—	—
Common qualified objects	—	—	—	—	—
All planned attempts	—	—	—	—	—

Table S35. **E29: same-state static-to-dynamic comparison (unmeasured).** Planned 2x2 study at an actually reached faulty state. Match target, inventory, candidate domain, images and remaining budget; retain separate total-budget sensitivity. Dashes mean not run, never zero.

Condition	Cases	SR	F1	Moves	Queries
Fresh solver / scheduled views	—	—	—	—	—
Fresh solver / active views	—	—	—	—	—
Continuing agent / scheduled views	—	—	—	—	—
Continuing agent / active views	—	—	—	—	—

Table S36. **E30: progress and stopping (unmeasured).** Planned full-trace postprocessing. Mean and peak F1 use actual operable states; external edits split stall segments. Completion and episode length must accompany averages to avoid rewarding early termination. Dashes mean not run, never zero.

Model	Mean F1	Peak F1	Stall span	Wrong finish	Completed N
GPT-6.1 Sol	—	—	—	—	—
Claude Sonnet 5.5	—	—	—	—	—
Gemini 3.8 Flash	—	—	—	—	—
Claude Haiku 5.5	—	—	—	—	—
Qwen3.5-9B	—	—	—	—	—
Qwen3.5-27B	—	—	—	—	—
InternVL3.5-8B	—	—	—	—	—
Qwen2.5-VL-7B	—	—	—	—	—
Qwen2.5-VL-32B	—	—	—	—	—
Gemma 3-12B	—	—	—	—	—

Table S37. **E31: dynamic input representation (unmeasured).** Planned adapters. No initial scene or automatic refresh after a move. Qualify normal success within each modality; report common-qualified objects and matched-state transfer separately. Dashes mean not run, never zero.

Input	D0 BCR	D1 NQR	D2 NQR	Queries	Eligible N
T: requested text state	—	—	—	—	—
I: requested scene images	—	—	—	—	—
TI: requested both	—	—	—	—	—
T: requested matched ledger	—	—	—	—	—
I: requested fact cards	—	—	—	—	—
TI: requested matched facts	—	—	—	—	—

Table S38. **E32: separate evidence and action demands (unmeasured).** Planned separate checkpoint probes. Answers are not injected into primary episodes; assistance changes the contract and does not prove a unique internal cause. Dashes mean not run, never zero.

Probe	Cases	Correct	Repair SR	Damage	Moves
Describe current vs target	—	—	—	—	—
Localize the affected part	—	—	—	—	—
Select a legal pose	—	—	—	—	—
Select access order	—	—	—	—	—
Execute a supplied feasible plan	—	—	—	—	—
Update after a hidden edit	—	—	—	—	—
Judge whether to finish	—	—	—	—	—

K. Complete Runnable Object Inventory

The gallery below contains every runnable configuration, followed by its stable ID, scope, active-part count, and question counts. Native release overview images retain the source context; the interactive review page additionally exposes focused current/target pairs, three angles, question-specific GT, and deep links. A tiny thumbnail is an inventory aid, not the image used for judging a defect. All 310 IDs are present in the current object index. Source attribution, original titles, and licenses remain in the machine-readable catalogue.

Current object inventory 1–40 / 310



Figure S4. Runnable object inventory, panel 1 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 41–80 / 310



Figure S5. Runnable object inventory, panel 2 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 81–120 / 310



Figure S6. Runnable object inventory, panel 3 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 121–160 / 310

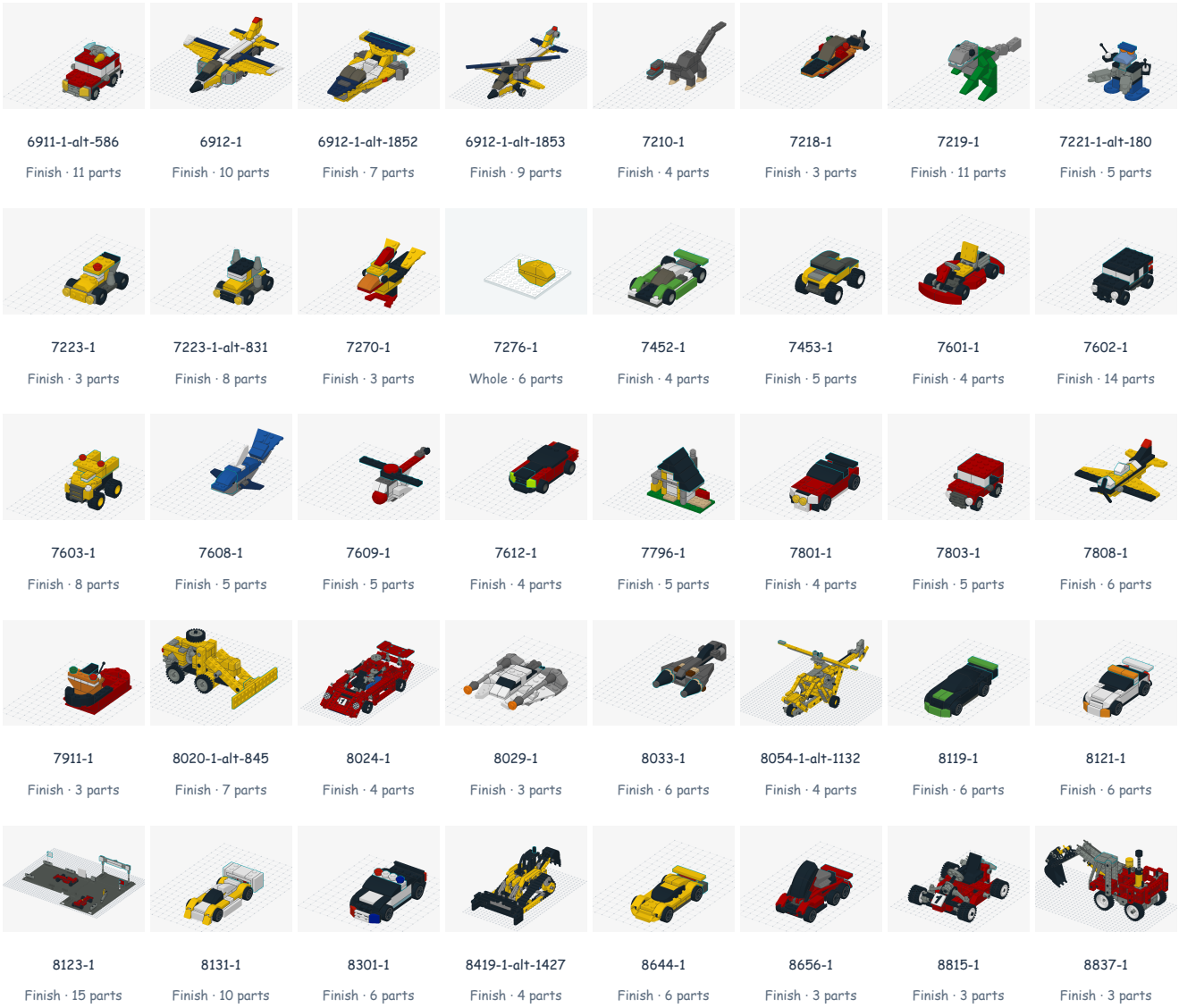


Figure S7. Runnable object inventory, panel 4 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 161–200 / 310



Figure S8. Runnable object inventory, panel 5 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 201–240 / 310

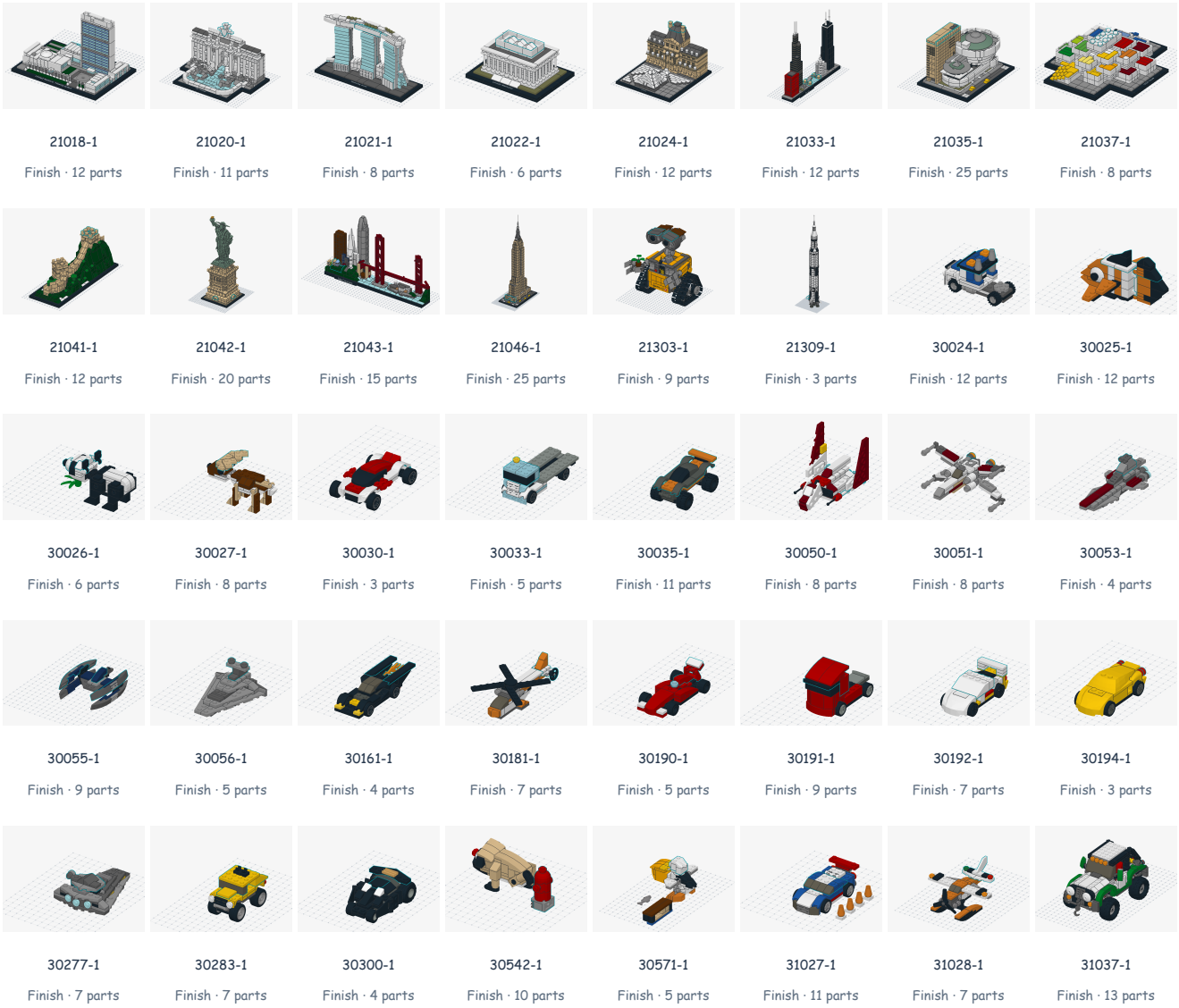


Figure S9. Runnable object inventory, panel 6 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 241–280 / 310

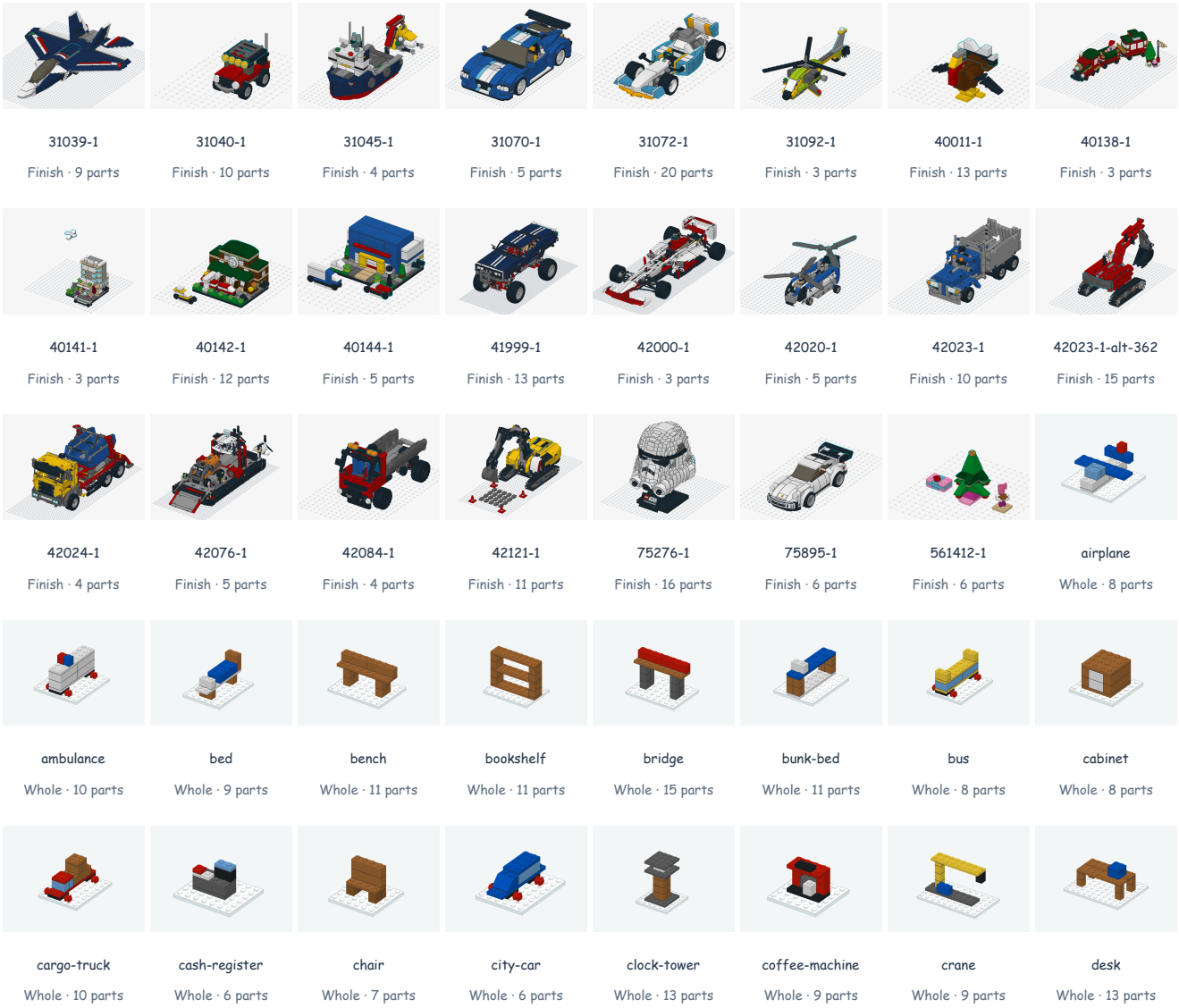


Figure S10. Runnable object inventory, panel 7 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Current object inventory 281–310 / 310

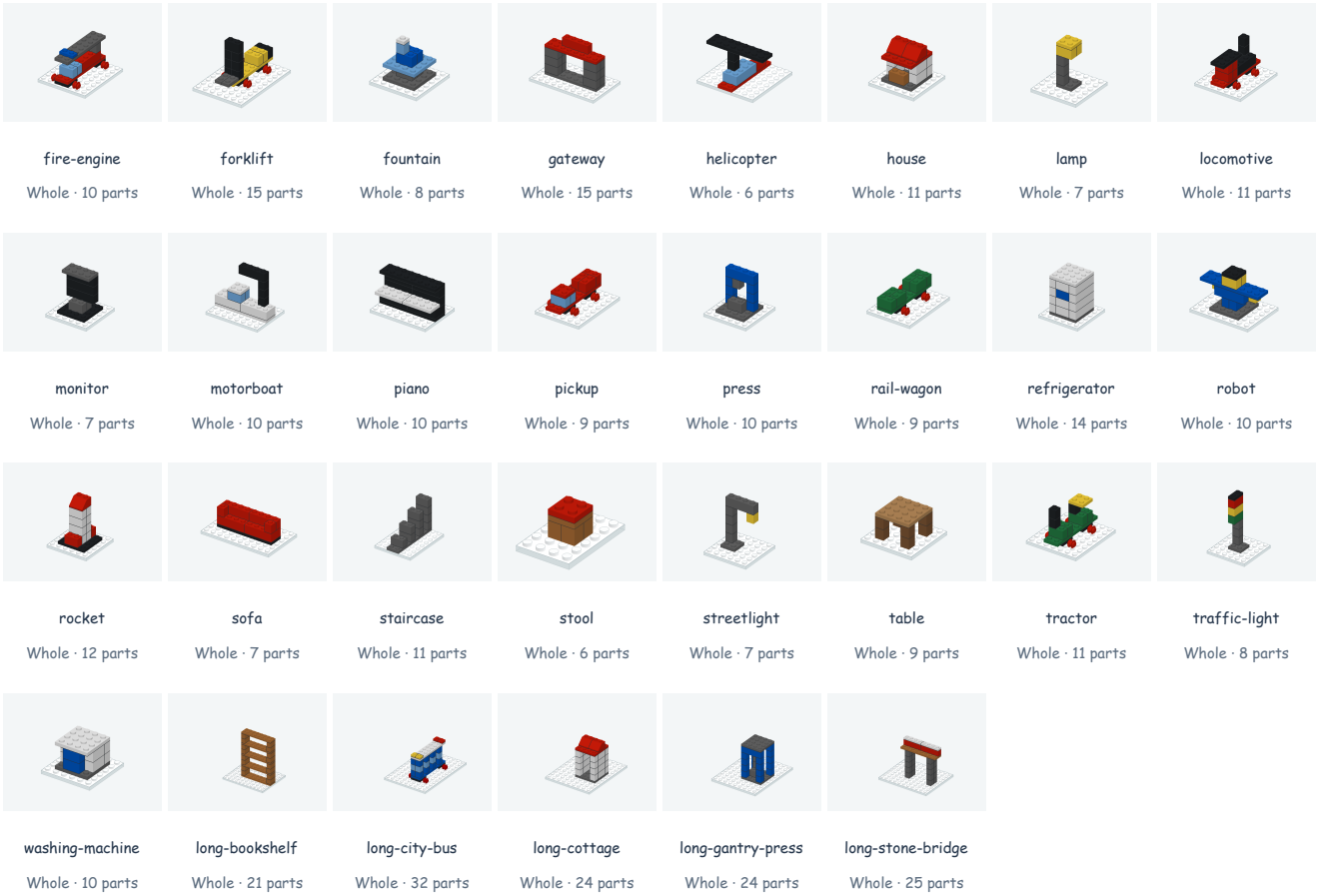


Figure S11. Runnable object inventory, panel 8 of 8. Whole and Finish denote task scope; part counts refer to the operable target.

Table S39. All 310 runnable object IDs. W: complete build; F: finishing task. S/M/L are length strata.

Object ID	Scope	Length	Parts	Static	Dynamic
sc-112-2	F	S	6	6	6
sc-126-1	F	M	15	6	7
sc-158-1	W	M	8	6	7
sc-311-1	F	S	5	6	6
sc-345-1	F	S	4	3	3
sc-390-2	F	M	12	6	7
sc-392-1	F	M	15	6	7
sc-544-1	F	M	8	6	7
sc-544-1-alt-1808	W	L	18	6	7
sc-544-1-alt-1809	F	M	13	6	7
sc-600-1	F	M	10	6	7
sc-601-2	F	S	4	6	6
sc-733-1-alt-1822	F	S	3	2	2
sc-733-1-alt-1823	F	S	4	6	6
sc-733-1-alt-1824	F	M	9	6	7
sc-850-1	F	S	3	6	6
sc-850-1-alt-102	F	M	8	6	7
sc-850-1-alt-103	F	S	5	6	6
sc-852-1	F	S	7	6	6
sc-1591-1	F	S	7	6	6
sc-1632-1	F	S	4	6	6
sc-1869-1	F	M	8	6	7
sc-2130-1	W	S	7	6	6
sc-2132-1	W	M	10	6	7
sc-2133-1	W	M	12	6	7
sc-3223-1	F	M	8	6	7
sc-3741-1	F	S	5	6	6
sc-4016-1	F	S	5	6	6
sc-4017-1	W	M	13	6	7
sc-4018-1	W	M	12	6	7
sc-4019-1	F	S	6	6	6
sc-4037-1	W	M	9	6	7
sc-4096-1-alt-1481	F	M	12	6	7
sc-4096-1-alt-1482	F	M	8	5	6
sc-4096-1-alt-1484	F	S	5	6	6
sc-4096-1-alt-1485	F	M	8	6	7
sc-4096-1-alt-1486	F	M	9	6	7
sc-4096-1-alt-1487	F	M	11	6	7
sc-4096-1-alt-1488	F	S	3	2	2
sc-4096-1-alt-1489	F	S	6	6	6
sc-4347-1	F	S	3	6	6
sc-4347-1-alt-794	F	S	5	6	6
sc-4348-1	F	S	6	6	6
sc-4404-1	F	S	4	6	6
sc-4487-1	F	S	4	4	4
sc-4488-1	F	S	3	5	5
sc-4489-1	F	M	11	6	7
sc-4491-1	F	M	9	6	7
sc-4492-1	F	M	13	6	7
sc-4495-1	F	S	3	4	4
sc-4536-1	F	M	8	6	7
sc-4837-1	F	M	13	6	7
sc-4837-1-alt-522	F	S	6	5	5
sc-4837-1-alt-526	F	M	12	6	7

Object ID	Scope	Length	Parts	Static	Dynamic
sc-4838-1	F	S	7	6	6
sc-4838-1-alt-23	F	S	4	6	6
sc-4838-1-alt-25	F	S	7	6	6
sc-4883-1-alt-1332	F	M	9	6	7
sc-4883-1-alt-1333	F	S	5	6	6
sc-4883-1-alt-1334	F	S	7	6	6
sc-4883-1-alt-1335	F	M	12	6	7
sc-4883-1-alt-1341	F	M	8	5	6
sc-4886-1	F	L	18	6	7
sc-4886-1-alt-800	F	M	14	6	7
sc-4886-1-alt-801	F	S	3	6	6
sc-4888-1-alt-803	F	S	3	6	6
sc-4896-1-alt-1349	F	S	6	6	6
sc-4903-1	F	M	9	6	7
sc-4904-1	F	S	6	6	6
sc-4905-1	F	S	3	5	5
sc-4911-1	F	S	3	4	4
sc-4914-1	F	S	4	6	6
sc-4915-1	F	S	5	6	6
sc-4917-1	F	S	7	6	6
sc-4918-1	F	M	11	6	7
sc-4918-1-alt-98	F	M	8	6	7
sc-4918-1-alt-100	F	M	12	6	7
sc-4939-1	F	S	3	5	5
sc-4939-1-alt-1714	F	M	10	6	7
sc-4948-1	F	S	3	4	4
sc-4953-1	F	S	6	6	6
sc-4953-1-alt-1719	F	M	10	6	7
sc-4954-1	F	S	7	6	6
sc-4954-1-alt-1355	F	L	17	6	7
sc-4954-1-alt-1356	F	M	10	6	7
sc-4956-1	F	M	10	6	7
sc-4956-1-alt-1358	F	L	25	6	7
sc-4956-1-alt-1359	F	L	26	6	7
sc-4993-1	F	S	7	6	6
sc-4993-1-alt-1361	F	S	7	6	6
sc-4996-1	F	M	10	6	7
sc-4996-1-alt-1364	F	M	11	6	7
sc-5521-1	F	S	4	5	5
sc-5571-1	F	S	6	6	6
sc-5761-1	F	S	5	6	6
sc-5762-1	F	M	12	6	7
sc-5762-1-alt-524	F	M	12	6	7
sc-5762-1-alt-525	F	M	8	6	7
sc-5764-1	F	S	4	6	6
sc-5765-1	F	M	14	6	7
sc-5864-1	F	S	5	6	6
sc-5864-1-alt-570	F	M	11	6	7
sc-5865-1	F	M	13	6	7
sc-5865-1-alt-1369	F	M	12	6	7
sc-5865-1-alt-1370	F	M	9	6	7
sc-5866-1	F	M	10	6	7
sc-5866-1-alt-1716	F	S	4	5	5
sc-5866-1-alt-1717	F	S	6	6	6
sc-5867-1	F	M	12	6	6
sc-5891-1	F	M	13	6	7

Object ID	Scope	Length	Parts	Static	Dynamic
sc-5891-1-alt-1372	F	M	11	6	7
sc-5891-1-alt-1373	F	M	12	6	7
sc-6742-1	F	S	6	6	6
sc-6742-1-alt-582	F	M	15	6	7
sc-6742-1-alt-583	F	M	12	6	7
sc-6743-1	F	S	5	6	6
sc-6745-1	F	S	7	6	6
sc-6753-1	F	M	12	6	7
sc-6911-1	F	M	11	6	7
sc-6911-1-alt-585	W	L	41	6	7
sc-6911-1-alt-586	F	M	11	6	7
sc-6912-1	F	M	10	6	7
sc-6912-1-alt-1852	F	S	7	5	5
sc-6912-1-alt-1853	F	M	9	6	7
sc-7210-1	F	S	4	6	6
sc-7218-1	F	S	3	6	6
sc-7219-1	F	M	11	6	7
sc-7221-1-alt-180	F	S	5	6	6
sc-7223-1	F	S	3	6	6
sc-7223-1-alt-831	F	M	8	6	7
sc-7270-1	F	S	3	3	3
sc-7276-1	W	S	6	6	6
sc-7452-1	F	S	4	6	6
sc-7453-1	F	S	5	6	6
sc-7601-1	F	S	4	3	3
sc-7602-1	F	M	14	6	7
sc-7603-1	F	M	8	6	7
sc-7608-1	F	S	5	6	6
sc-7609-1	F	S	5	6	6
sc-7612-1	F	S	4	5	5
sc-7796-1	F	S	5	6	6
sc-7801-1	F	S	4	6	6
sc-7803-1	F	S	5	6	6
sc-7808-1	F	S	6	6	6
sc-7911-1	F	S	3	6	6
sc-8020-1-alt-845	F	S	7	6	6
sc-8024-1	F	S	4	4	4
sc-8029-1	F	S	3	6	6
sc-8033-1	F	S	6	4	4
sc-8054-1-alt-1132	F	S	4	6	6
sc-8119-1	F	S	6	6	6
sc-8121-1	F	S	6	6	6
sc-8123-1	F	M	15	6	7
sc-8131-1	F	M	10	6	7
sc-8301-1	F	S	6	6	6
sc-8419-1-alt-1427	F	S	4	2	2
sc-8644-1	F	S	6	6	6
sc-8656-1	F	S	3	5	5
sc-8815-1	F	S	3	3	3
sc-8837-1	F	S	3	6	6
sc-8837-1-alt-1857	F	S	5	6	6
sc-9391-1	F	S	4	3	3
sc-9391-1-alt-987	F	M	8	4	5
sc-9394-1-alt-991	F	S	3	5	5
sc-9395-1	F	M	14	6	7
sc-9397-1	F	M	8	4	4

Object ID	Scope	Length	Parts	Static	Dynamic
sc-10013-1	F	S	6	5	5
sc-10014-1	F	M	8	6	7
sc-10015-1	F	M	13	5	6
sc-10016-1	F	S	5	6	6
sc-10017-1	F	M	8	5	6
sc-10124-1	F	M	10	6	7
sc-10170-1	F	M	14	6	7
sc-10175-1	F	M	8	3	4
sc-10205-1	F	S	5	6	6
sc-10226-1	F	M	15	6	7
sc-10241-1	F	S	3	2	2
sc-10242-1	F	S	4	3	3
sc-10248-1	F	M	13	5	6
sc-10258-1	F	L	17	6	7
sc-10262-1	F	L	16	6	6
sc-10265-1	F	L	24	6	7
sc-10267-1	F	M	13	6	6
sc-10315-1	F	S	4	2	2
sc-20006-1	F	S	3	5	5
sc-20011-1	F	M	8	6	7
sc-20015-1	F	S	3	6	6
sc-20019-1	F	S	7	6	6
sc-21000-1	F	M	11	6	7
sc-21001-1	F	M	13	5	6
sc-21003-1	F	S	3	2	2
sc-21005-1	F	M	10	6	7
sc-21009-1	F	S	5	6	6
sc-21010-1	F	S	4	5	5
sc-21011-1	F	M	10	6	7
sc-21012-1	F	S	3	4	4
sc-21013-1	F	M	15	6	7
sc-21014-1	F	S	5	6	6
sc-21016-1	F	M	8	6	7
sc-21017-1	F	L	24	6	7
sc-21018-1	F	M	12	6	7
sc-21020-1	F	M	11	6	7
sc-21021-1	F	M	8	6	7
sc-21022-1	F	S	6	6	6
sc-21024-1	F	M	12	6	7
sc-21033-1	F	M	12	6	7
sc-21035-1	F	L	25	6	7
sc-21037-1	F	M	8	4	5
sc-21041-1	F	M	12	6	7
sc-21042-1	F	L	20	6	7
sc-21043-1	F	M	15	6	7
sc-21046-1	F	L	25	6	7
sc-21303-1	F	M	9	6	7
sc-21309-1	F	S	3	5	5
sc-30024-1	F	M	12	6	7
sc-30025-1	F	M	12	6	7
sc-30026-1	F	S	6	6	6
sc-30027-1	F	M	8	6	7
sc-30030-1	F	S	3	5	5
sc-30033-1	F	S	5	6	6
sc-30035-1	F	M	11	6	7
sc-30050-1	F	M	8	5	6

Object ID	Scope	Length	Parts	Static	Dynamic
sc-30051-1	F	M	8	6	7
sc-30053-1	F	S	4	6	6
sc-30055-1	F	M	9	6	7
sc-30056-1	F	S	5	6	6
sc-30161-1	F	S	4	6	6
sc-30181-1	F	S	7	6	6
sc-30190-1	F	S	5	6	6
sc-30191-1	F	M	9	6	6
sc-30192-1	F	S	7	6	6
sc-30194-1	F	S	3	3	3
sc-30277-1	F	S	7	6	6
sc-30283-1	F	S	7	6	6
sc-30300-1	F	S	4	6	6
sc-30542-1	F	M	10	5	6
sc-30571-1	F	S	5	6	6
sc-31027-1	F	M	11	6	7
sc-31028-1	F	S	7	6	6
sc-31037-1	F	M	13	6	7
sc-31039-1	F	M	9	6	7
sc-31040-1	F	M	10	6	7
sc-31045-1	F	S	4	6	6
sc-31070-1	F	S	5	6	6
sc-31072-1	F	L	20	6	7
sc-31092-1	F	S	3	2	2
sc-40011-1	F	M	13	6	7
sc-40138-1	F	S	3	4	4
sc-40141-1	F	S	3	5	5
sc-40142-1	F	M	12	6	7
sc-40144-1	F	S	5	5	5
sc-41999-1	F	M	13	6	7
sc-42000-1	F	S	3	6	6
sc-42020-1	F	S	5	6	6
sc-42023-1	F	M	10	6	7
sc-42023-1-alt-362	F	M	15	6	7
sc-42024-1	F	S	4	5	5
sc-42076-1	F	S	5	6	6
sc-42084-1	F	S	4	6	6
sc-42121-1	F	M	11	6	7
sc-75276-1	F	L	16	6	7
sc-75895-1	F	S	6	6	6
sc-561412-1	F	S	6	6	6
obj-airplane	W	M	8	0	38
obj-ambulance	W	M	10	0	25
obj-bed	W	M	9	0	38
obj-bench	W	M	11	0	32
obj-bookshelf	W	M	11	0	47
obj-bridge	W	M	15	0	28
obj-bunk-bed	W	M	11	0	23
obj-bus	W	M	8	0	29
obj-cabinet	W	M	8	0	45
obj-cargo-truck	W	M	10	0	41
obj-cash-register	W	S	6	0	19
obj-chair	W	S	7	0	20
obj-city-car	W	S	6	0	21
obj-clock-tower	W	M	13	0	46
obj-coffee-machine	W	M	9	0	28

Object ID	Scope	Length	Parts	Static	Dynamic
obj-crane	W	M	9	0	23
obj-desk	W	M	13	0	40
obj-fire-engine	W	M	10	0	35
obj-forklift	W	M	15	0	39
obj-fountain	W	M	8	0	32
obj-gateway	W	M	15	0	36
obj-helicopter	W	S	6	0	17
obj-house	W	M	11	0	41
obj-lamp	W	S	7	0	14
obj-locomotive	W	M	11	0	36
obj-monitor	W	S	7	0	20
obj-motorboat	W	M	10	0	23
obj-piano	W	M	10	0	40
obj-pickup	W	M	9	0	37
obj-press	W	M	10	0	45
obj-rail-wagon	W	M	9	0	37
obj-refrigerator	W	M	14	0	45
obj-robot	W	M	10	0	38
obj-rocket	W	M	12	0	57
obj-sofa	W	S	7	0	22
obj-staircase	W	M	11	0	40
obj-stool	W	S	6	0	17
obj-streetlight	W	S	7	0	13
obj-table	W	M	9	0	31
obj-tractor	W	M	11	0	35
obj-traffic-light	W	M	8	0	23
obj-washing-machine	W	M	10	0	50
v7-long-bookshelf	W	L	21	24	23
v7-long-city-bus	W	L	32	24	23
v7-long-cottage	W	L	24	22	23
v7-long-gantry-press	W	L	24	26	25
v7-long-stone-bridge	W	L	25	26	23